# Sam Gunn


CONTACT

Massachusetts Institute of Technology
Building 2, Room 246A
Cambridge, MA 02139 USA

Email: gunn@mit.edu
Website: samuel-gunn.github.io


POSITIONS

**Massachusetts Institute of Technology**
C.L.E. Moore Instructor of Mathematics (Summer 2026 to Present)

**Institute for Responsible Superintelligence (RESI)**
Researcher (September 2026 to Present)

**Google Research, New York, NY**
Student Researcher (January 2025 to July 2025); host: Pritish Kamath

**Google Quantum AI, Los Angeles, CA**
Student Researcher (June 2024 to September 2024); host: Ramis Movassagh

EDUCATION

**The University of California, Berkeley**
Ph.D. in Computer Science (Fall 2020 to Present; expected Fall 2026)
Advisor: Umesh Vazirani

**The University of Texas at Austin**
B.S. in Mathematics (Fall 2016 to Spring 2020)
Advisors: Scott Aaronson and Cameron Gordon

RESEARCH

*Conference publications*

1. Sam Gunn. "How to sketch a learning algorithm." *18th Innovations in Theoretical Computer Science Conference (ITCS)*. 2027. To appear. arXiv: 2604.07328.
2. Miranda Christ, Noah Golowich, Sam Gunn, Ankur Moitra, and Daniel Wichs. "Improved pseudorandom codes from permuted puzzles." *Proceedings of the 58th Annual ACM Symposium on Theory of Computing (STOC)*. 2026. DOI: 10.1145/3798129.3800916. ePrint: 2025/2222.
3. Sanjam Garg, Sam Gunn, and Mingyuan Wang. "Black-box crypto is useless for pseudorandom codes." *Proceedings of the 23rd Theory of Cryptography Conference (TCC)*. 2025. DOI: 10.1007/978-3-032-12290-2_7. ePrint: 2025/1021. Outstanding Paper Award.
4. Sam Gunn, Xuandong Zhao, and Dawn Song. "An undetectable watermark for generative image models." *The Thirteenth International Conference on Learning Representations (ICLR)*. 2025. arXiv: 2410.07369.
5. Xuandong Zhao, Sam Gunn, Miranda Christ, Jaiden Fairoze, Andres Fabrega, Nicholas Carlini, Sanjam Garg, Sanghyun Hong, Milad Nasr, Florian Tramer, Somesh Jha, Lei Li, Yu-Xiang Wang, and Dawn Song. "SoK: Watermarking for AI-generated content." *Proceedings of the 46th IEEE Symposium on Security and Privacy (S&P)*. 2025. DOI: 10.1109/SP61157.2025.00178. arXiv: 2411.18479.
6. Omar Alrabiah, Prabhanjan Ananth, Miranda Christ, Yevgeniy Dodis, and Sam Gunn. "Ideal pseudorandom codes." *Proceedings of the 57th Annual ACM Symposium on Theory of Computing (STOC)*. 2025. DOI: 10.1145/3717823.3718309. ePrint: 2024/1840. Invited to the SICOMP Special Issue for STOC 2025.
7. Sam Gunn and Ramis Movassagh. "Quantum one-time protection of any randomized algorithm." *Proceedings of the 45th Annual International Cryptology Conference (CRYPTO)*. 2025. DOI: 10.1007/978-3-032-01878-6_11. ePrint: 2024/1798.
8. Sam Gunn, Yael Tauman Kalai, Anand Natarajan, and Ági Villányi. "Classical commitments to quantum states." *Proceedings of the 57th Annual ACM Symposium on Theory of Computing (STOC)*. 2025. DOI: 10.1145/3717823.3718264. ePrint: 2024/606. Presented at *Quantum Information Processing (QIP 2025)*.

9. Miranda Christ and Sam Gunn. "Pseudorandom error-correcting codes." *Proceedings of the 44th Annual International Cryptology Conference (CRYPTO)*. 2024. DOI: 10.1007/978-3-031-68391-6_10. ePrint: 2024/235.
10. Miranda Christ, Sam Gunn, and Or Zamir. "Undetectable watermarks for language models." *Proceedings of the 37th Annual Conference on Learning Theory (COLT)*. 2024. ePrint: 2023/763. Presented at *Real World Crypto (RWC 2024)*.
11. Andrea Coladangelo and Sam Gunn. "How to use quantum indistinguishability obfuscation." *Proceedings of the 56th Annual ACM Symposium on Theory of Computing (STOC)*. 2024. DOI: 10.1145/3618260.3649779. ePrint: 2023/1756.
12. Thiago Bergamaschi, Louis Golowich, and Sam Gunn. "Approaching the quantum Singleton bound with approximate error correction." *Proceedings of the 56th Annual ACM Symposium on Theory of Computing (STOC)*. 2024. DOI: 10.1145/3618260.3649680. arXiv: 2212.09935. Presented at *Quantum Information Processing (QIP 2024)*.
13. Sam Gunn, Nathan Ju, Fermi Ma, and Mark Zhandry. "Commitments to quantum states." *Proceedings of the 55th Annual ACM Symposium on Theory of Computing (STOC)*. 2023. DOI: 10.1145/3564246.3585198. ePrint: 2022/1358. Presented at *Quantum Information Processing (QIP 2023)*.
14. Sam Gunn, Doseok Jang, Orr Paradise, Lucas Spangher, and Costas J. Spanos. "Adversarial poisoning attacks on reinforcement learning-driven energy pricing." *Proceedings of the 9th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys)*. 2022. DOI: 10.1145/3563357.3564075.

*Journal publications*

1. Roozbeh Bassirian, Adam Bouland, Bill Fefferman, Sam Gunn, and Avishay Tal. "On certified randomness from Fourier sampling or random circuit sampling." *Quantum* 10, 2002 (2026). DOI: 10.22331/q-2026-02-10-2002. arXiv: 2111.14846.
2. Miranda Christ, Sam Gunn, Tal Malkin, and Mariana Raykova. "Provably robust watermarks for open-source language models." *Transactions on Machine Learning Research* (2026).
3. Sam Gunn, Nathan Ju, Fermi Ma, and Mark Zhandry. "Commitments to quantum states." *SIAM Journal on Computing*, Special Issue for the 55th Annual ACM Symposium on Theory of Computing (STOC). In press. ePrint: 2022/1358.
4. Scott Aaronson and Sam Gunn. "On the classical hardness of spoofing linear cross-entropy benchmarking." *Theory of Computing* 16, Article 11, pp. 1–8 (2020). DOI: 10.4086/toc.2020.v016a011. Presented at the *Twenty-Second Annual Southwest Quantum Information and Technology Workshop (SQuInT 2020)*.

## Coverage

- Samuel Greengard. "Can Generative Watermarking Make a Mark?" *Communications of the ACM*, December 29, 2025.
- Anil Ananthaswamy. "Watermarks and Pseudorandom Codes." *Polynomial Times*, Simons Institute, 2025–26, Issue 1.
- Teresa Brooks-Mejia and Christopher Patton. "An early look at cryptographic watermarks for AI-generated content." Cloudflare Blog, March 19, 2025.

## Awards and funding

- Lambda Research Grant, 2026
- TCC Outstanding Paper Award, 2025
- Google PhD Fellowship, 2022–2025
- UC Berkeley Lotfi A. Zadeh Prize, 2024–2025
- UC Berkeley EECS Department Merit Award, 2024
- UC Berkeley EECS Departmental Fellowship, Fall 2020

## TEACHING

- MIT, 18.02: Calculus. Recitation Instructor (Fall 2026).
- UC Berkeley, CS 172: Computability and complexity. Graduate Student Instructor (Fall 2024).
- UC Berkeley, CS 170: Efficient algorithms and intractable problems. Graduate Student Instructor (Spring 2023).

## TALKS

"Tracing data in AI"

- EPFL Confluence Workshop — October 2026

"How to sketch a learning algorithm"

- NYU Center for Data Science Seminar — October 2026
- MIT Statistics and Data Science Seminar — September 2026
- Invited talk at ITC 2026 — August 2026
- Flatiron Institute — June 2026
- The Role of Theory in Trustworthy & Interpretable AI, STOC 2026 — June 2026
- MIT ML+Cryptography Seminar — May 2026
- Institute for Advanced Study — May 2026

"On undetectable backdoors"

- UT Austin Theory and Alignment Group — November 2025

"Pseudorandom error-correcting codes"

- TCC 2025 — December 2025
- Harvard Theory Seminar — September 2025
- NYU Cryptography Seminar — April 2025
- CRYPTO 2024 — August 2024
- UCSB Theory Colloquium — June 2024
- Berkeley Theory Lunch — April 2024
- Google Research NYC — April 2024
- NTT Research — March 2024

"Undetectable watermarks for language models"

- NYU Cryptography Seminar — September 2023
- Berkeley ML Theory Seminar — September 2023
- MIT Cryptography Group — August 2023
- Google Research Mountain View — July 2023

"Quantum one-time protection of any randomized algorithm"

- CRYPTO 2025 — August 2025

"Quantum indistinguishability obfuscation"

- University of Ottawa Quantum Seminar — April 2024
- CIQC Spark Talk — March 2024

"Approaching the quantum Singleton bound with approximate error correction"

- Simons Quantum Colloquium — December 2022

"Commitments to quantum states"

- STOC 2023 — June 2023

"Unclonable cryptography"

- Guest lecture for Quantum Complexity, UC Berkeley — December 2022

"Certified entropy from random circuit sampling"

- Berkeley Theory Lunch — March 2022

## SERVICE

Program Committee, ITCS — 2027
MIT Computer Science UROP Mentor — Summer 2026
Berkeley Mathematics DRP Mentor — Spring 2021

## REVIEWING

STOC, FOCS, SODA, CRYPTO, EUROCRYPT, TCC, NeurIPS, ICLR, QIP, Annals of Statistics